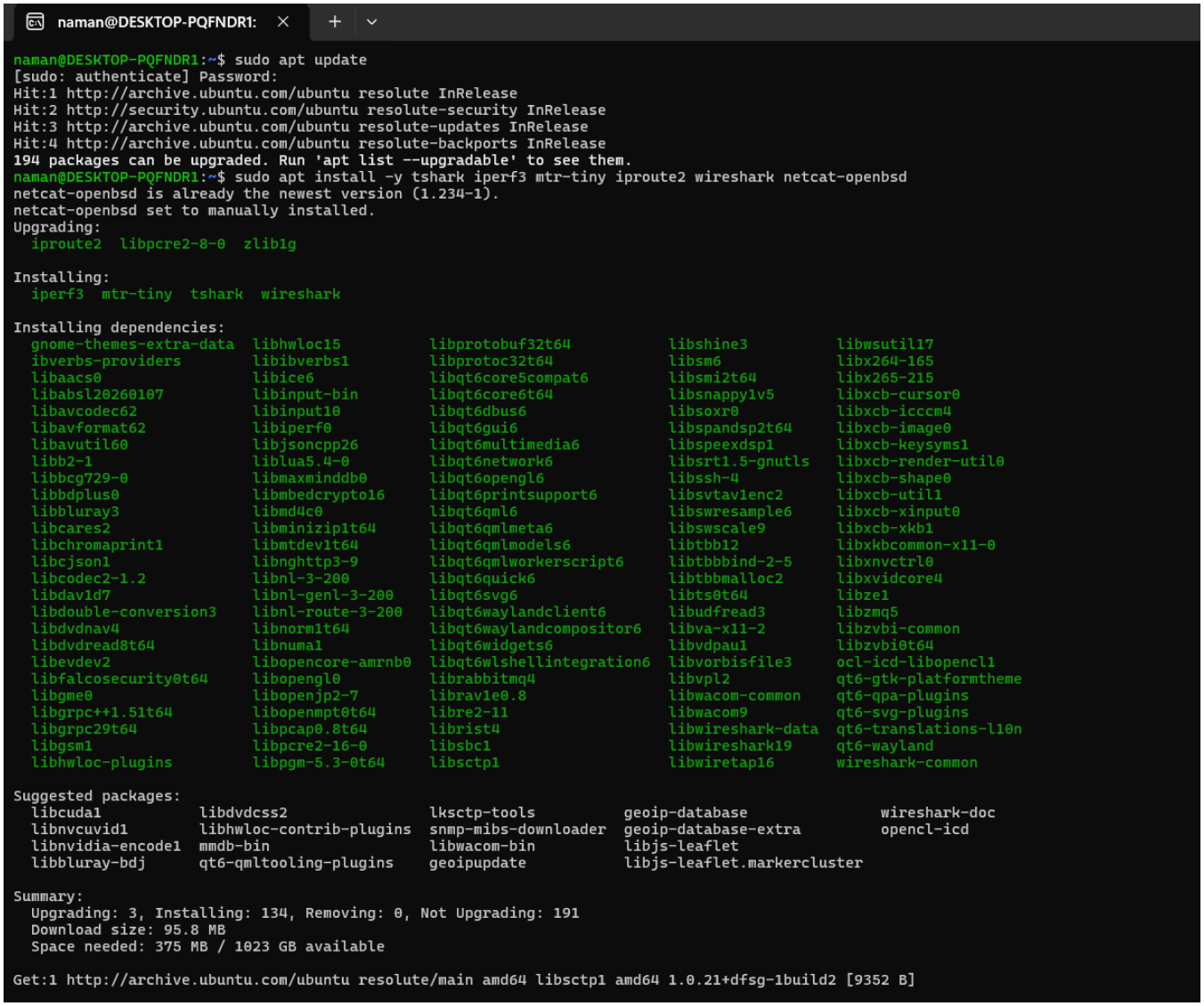
CNNP SKILL 2

Name: Naman Masineni Roll Number: 2520090194

Install required tools – apt update and installation



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Get:2 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libiperf0 amd64 3.20-2 [93.2 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/universe amd64 iperf3 amd64 3.20-2 [19.9 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libpcre2-8-0 amd64 10.46-1ubuntu0.1 [257 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 zlib1g amd64 1:1.3.dfsg+really1.3.1-1ubuntu3.2 [61.8 kB]
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Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libnl-3-200 amd64 3.12.0-2 [58.8 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 libnl-route-3-200 amd64 3.12.0-2 [204 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/main amd64 libibverbs1 amd64 61.0-2ubuntu3 [70.6 kB]
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Get:11 http://archive.ubuntu.com/ubuntu resolute/main amd64 libevdev2 amd64 1.13.6+dfsg-1 [37.7 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute/main amd64 libmaxminddb0 amd64 1.12.2-1build2 [18.9 kB]
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Get:137 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libhwloc-plugins amd64 2.13.0-2 [21.5 kB]
Fetched 95.8 MB in 24s (3960 kB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package libsctp1:amd64.
(Reading database ... 48243 files and directories currently installed.)
Preparing to unpack .../libsctp1_1.0.21+dfsg-1build2_amd64.deb ...
Unpacking libsctp1:amd64 (1.0.21+dfsg-1build2) ...
Selecting previously unselected package libiperf0:amd64.
Preparing to unpack .../libiperf0_3.20-2_amd64.deb ...
Unpacking libiperf0:amd64 (3.20-2) ...
Selecting previously unselected package iperf3.
Preparing to unpack .../iperf3_3.20-2_amd64.deb ...
Unpacking iperf3 (3.20-2) ...
Preparing to unpack .../libpcre2-8-0_10.46-1ubuntu0.1_amd64.deb ...
Unpacking libpcre2-8-0:amd64 (10.46-1ubuntu0.1) over (10.46-1build1) ...
Setting up libpcre2-8-0:amd64 (10.46-1ubuntu0.1) ...
(Reading database ... 48265 files and directories currently installed.)
Preparing to unpack .../zlib1g_1%3a1.3.dfsg+really1.3.1-1ubuntu3.2_amd64.deb ...
Unpacking zlib1g:amd64 (1:1.3.dfsg+really1.3.1-1ubuntu3.2) over (1:1.3.dfsg+really1.3.1-1ubuntu3) ...
Setting up zlib1g:amd64 (1:1.3.dfsg+really1.3.1-1ubuntu3.2) ...
(Reading database ... 48265 files and directories currently installed.)
Preparing to unpack .../000-iproute2_6.19.0-1ubuntu1.1_amd64.deb ...
Unpacking iproute2 (6.19.0-1ubuntu1.1) over (6.19.0-1ubuntu1) ...
Selecting previously unselected package libnl-3-200:amd64.
Preparing to unpack .../001-libnl-3-200_3.12.0-2_amd64.deb ...
Unpacking libnl-3-200:amd64 (3.12.0-2) ...
Selecting previously unselected package libnl-route-3-200:amd64.
Preparing to unpack .../002-libnl-route-3-200_3.12.0-2_amd64.deb ...
Unpacking libnl-route-3-200:amd64 (3.12.0-2) ...
Selecting previously unselected package libibverbs1:amd64.
Preparing to unpack .../003-libibverbs1_61.0-2ubuntu3_amd64.deb ...
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Unpacking libibverbs1:amd64 (61.0-2ubuntu3) ...
Selecting previously unselected package ibverbs-providers:amd64.
Preparing to unpack .../004-ibverbs-providers_61.0-2ubuntu3_amd64.deb ...
Unpacking ibverbs-providers:amd64 (61.0-2ubuntu3) ...
Selecting previously unselected package libevdev2:amd64.
Preparing to unpack .../005-libevdev2_1.13.6+dfsg-1_amd64.deb ...
Unpacking libevdev2:amd64 (1.13.6+dfsg-1) ...
Selecting previously unselected package libmaxminddb0:amd64.
Preparing to unpack .../006-libmaxminddb0_1.12.2-1build2_amd64.deb ...
Unpacking libmaxminddb0:amd64 (1.12.2-1build2) ...
Selecting previously unselected package libnuma1:amd64.
Preparing to unpack .../007-libnuma1_2.0.19-1build1_amd64.deb ...
Unpacking libnuma1:amd64 (2.0.19-1build1) ...
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Preparing to unpack .../008-libpcap0.8t64_1.10.6-1ubuntu1.1_amd64.deb ...
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Unpacking mtr-tiny (0.95-1.1ubuntu2) ...
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Unpacking gnome-themes-extra-data (3.28-5) ...
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Preparing to unpack .../012-libabsl20260107_20260107.0-4_amd64.deb ...
Unpacking libabsl20260107:amd64 (20260107.0-4) ...
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Preparing to unpack .../013-libva-x11-2_2.23.0-1ubuntu1_amd64.deb ...
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Preparing to unpack .../014-libvdpau1_1.5-4_amd64.deb ...
Unpacking libvdpau1:amd64 (1.5-4) ...
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Preparing to unpack .../015-libvpl2_1%3a2.16.0-1_amd64.deb ...
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Selecting previously unselected package ocl-icd-libopencl1:amd64.
Preparing to unpack .../016-ocl-icd-libopencl1_2.3.4-1_amd64.deb ...
Unpacking ocl-icd-libopencl1:amd64 (2.3.4-1) ...
Selecting previously unselected package libavutil60:amd64.
Preparing to unpack .../017-libavutil60_7%3a8.0.1-3ubuntu2_amd64.deb ...
Unpacking libavutil60:amd64 (7:8.0.1-3ubuntu2) ...
Selecting previously unselected package libcodec2-1.2:amd64.
Preparing to unpack .../018-libcodec2-1.2_1.2.0-4_amd64.deb ...
Unpacking libcodec2-1.2:amd64 (1.2.0-4) ...
Selecting previously unselected package libdav1d7:amd64.
Preparing to unpack .../019-libdav1d7_1.5.3-1_amd64.deb ...
Unpacking libdav1d7:amd64 (1.5.3-1) ...
Selecting previously unselected package libgsm1:amd64.
Preparing to unpack .../020-libgsm1_1.0.23-2_amd64.deb ...
Unpacking libgsm1:amd64 (1.0.23-2) ...
Selecting previously unselected package libopenjp2-7:amd64.
Preparing to unpack .../021-libopenjp2-7_2.5.4-1ubuntu0.1_amd64.deb ...
Unpacking libopenjp2-7:amd64 (2.5.4-1ubuntu0.1) ...
Selecting previously unselected package librav1e0.8:amd64.
```

```
Unpacking libtbb12:amd64 (2022.3.0-2) ...
Selecting previously unselected package libfalcosecurity0t64:amd64.
Preparing to unpack .../067-libfalcosecurity0t64_0.20.0-4ubuntu0.1_amd64.deb ...
Unpacking libfalcosecurity0t64:amd64 (0.20.0-4ubuntu0.1) ...
Selecting previously unselected package libice6:amd64.
Preparing to unpack .../068-libice6_2%3a1.1.1-1build1_amd64.deb ...
Unpacking libice6:amd64 (2:1.1.1-1build1) ...
Selecting previously unselected package libwacom-common.
Preparing to unpack .../069-libwacom-common_2.18.0-1_all.deb ...
Unpacking libwacom-common (2.18.0-1) ...
Selecting previously unselected package libwacom9:amd64.
Preparing to unpack .../070-libwacom9_2.18.0-1_amd64.deb ...
Unpacking libwacom9:amd64 (2.18.0-1) ...
Selecting previously unselected package libinput-bin.
Preparing to unpack .../071-libinput-bin_1.31.1-1ubuntu1.2_amd64.deb ...
Unpacking libinput-bin (1.31.1-1ubuntu1.2) ...
Selecting previously unselected package libmtdev1t64:amd64.
Preparing to unpack .../072-libmtdev1t64_1.1.7-1build1_amd64.deb ...
Unpacking libmtdev1t64:amd64 (1.1.7-1build1) ...
Selecting previously unselected package libinput10:amd64.
Preparing to unpack .../073-libinput10_1.31.1-1ubuntu1.2_amd64.deb ...
Unpacking libinput10:amd64 (1.31.1-1ubuntu1.2) ...
Selecting previously unselected package liblua5.4-0:amd64.
Preparing to unpack .../074-liblua5.4-0_5.4.8-1build1_amd64.deb ...
Unpacking liblua5.4-0:amd64 (5.4.8-1build1) ...
Selecting previously unselected package libmd4c0:amd64.
Preparing to unpack .../075-libmd4c0_0.5.2-2build1_amd64.deb ...
Unpacking libmd4c0:amd64 (0.5.2-2build1) ...
Selecting previously unselected package libminizip1t64:amd64.
Preparing to unpack .../076-libminizip1t64_1%3a1.3.dfsg+really1.3.1-1ubuntu3.2_amd64.deb ...
Unpacking libminizip1t64:amd64 (1:1.3.dfsg+really1.3.1-1ubuntu3.2) ...
Selecting previously unselected package libnghttp3-9:amd64.
Preparing to unpack .../077-libnghttp3-9_1.12.0-1_amd64.deb ...
Unpacking libnghttp3-9:amd64 (1.12.0-1) ...
Selecting previously unselected package libnl-genl-3-200:amd64.
Preparing to unpack .../078-libnl-genl-3-200_3.12.0-2_amd64.deb ...
Unpacking libnl-genl-3-200:amd64 (3.12.0-2) ...
Selecting previously unselected package libopencore-amrnb0:amd64.
Preparing to unpack .../079-libopencore-amrnb0_0.1.6-1build2_amd64.deb ...
Unpacking libopencore-amrnb0:amd64 (0.1.6-1build2) ...
Selecting previously unselected package libpcre2-16-0:amd64.
Preparing to unpack .../080-libpcre2-16-0_10.46-1ubuntu0.1_amd64.deb ...
Unpacking libpcre2-16-0:amd64 (10.46-1ubuntu0.1) ...
Selecting previously unselected package libqt6core6t64:amd64.
Preparing to unpack .../081-libqt6core6t64_6.10.2+dfsg-7_amd64.deb ...
Unpacking libqt6core6t64:amd64 (6.10.2+dfsg-7) ...
Selecting previously unselected package libqt6core5compat6:amd64.
Preparing to unpack .../082-libqt6core5compat6_6.10.2-1_amd64.deb ...
Unpacking libqt6core5compat6:amd64 (6.10.2-1) ...
Selecting previously unselected package libqt6dbus6:amd64.
Preparing to unpack .../083-libqt6dbus6_6.10.2+dfsg-7_amd64.deb ...
Unpacking libqt6dbus6:amd64 (6.10.2+dfsg-7) ...
Selecting previously unselected package libopengl0:amd64.
Preparing to unpack .../084-libopengl0_1.7.0-3_amd64.deb ...
Unpacking libopengl0:amd64 (1.7.0-3) ...
Selecting previously unselected package libsm6:amd64.
```

```
Setting up libzmq5:amd64 (4.3.5-1build3) ...
Setting up libzvbi0t64:amd64 (0.2.44-1ubuntu2) ...
Setting up libavutil60:amd64 (7:8.0.1-3ubuntu2) ...
Setting up libwacom9:amd64 (2.18.0-1) ...
Setting up libwiretap16:amd64 (4.6.4-1) ...
Setting up libhwloc-plugins:amd64 (2.13.0-2) ...
Setting up libinput-bin (1.31.1-1ubuntu1.2) ...
Setting up libtbbbind-2-5:amd64 (2022.3.0-2) ...
Setting up libnl-route-3-200:amd64 (3.12.0-2) ...
Setting up librist4:amd64 (0.2.11+dfsg-1build1) ...
Setting up libswscale9:amd64 (7:8.0.1-3ubuntu2) ...
Setting up libgrpc29t64:amd64 (1.51.1-8ubuntu1) ...
Setting up libiperf0:amd64 (3.20-2) ...
Setting up libswresample6:amd64 (7:8.0.1-3ubuntu2) ...
Setting up libnl-genl-3-200:amd64 (3.12.0-2) ...
Setting up libqt6core5compat6:amd64 (6.10.2-1) ...
Setting up libqt6dbus6:amd64 (6.10.2+dfsg-7) ...
Setting up libinput10:amd64 (1.31.1-1ubuntu1.2) ...
Setting up libibverbs1:amd64 (61.0-2ubuntu3) ...
Setting up ibverbs-providers:amd64 (61.0-2ubuntu3) ...
Setting up iperf3 (3.20-2) ...
Setting up libgrpc++1.51t64:amd64 (1.51.1-8ubuntu1) ...
Setting up libtbb12:amd64 (2022.3.0-2) ...
Setting up libqt6gui6:amd64 (6.10.2+dfsg-7) ...
Setting up qt6-gtk-platformtheme:amd64 (6.10.2+dfsg-7) ...
Setting up libavcodec62:amd64 (7:8.0.1-3ubuntu2) ...
Setting up libfalcosecurity0t64:amd64 (0.20.0-4ubuntu0.1) ...
Setting up libqt6network6:amd64 (6.10.2+dfsg-7) ...
Setting up libpcap0.8t64:amd64 (1.10.6-1ubuntu1.1) ...
Setting up qt6-qpa-plugins:amd64 (6.10.2+dfsg-7) ...
Setting up libavformat62:amd64 (7:8.0.1-3ubuntu2) ...
Setting up libqt6qml6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6opengl6:amd64 (6.10.2+dfsg-7) ...
Setting up libqt6qmlmodels6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6widgets6:amd64 (6.10.2+dfsg-7) ...
Setting up libqt6qmlworkerscript6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6svg6:amd64 (6.10.2-2) ...
Setting up libwireshark19:amd64 (4.6.4-1) ...
Setting up libqt6waylandclient6:amd64 (6.10.2+dfsg-7) ...
Setting up qt6-svg-plugins:amd64 (6.10.2-2) ...
Setting up libqt6qmlmeta6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6wlshellintegration6:amd64 (6.10.2+dfsg-7) ...
Setting up wireshark-common (4.6.4-1) ...
Setting up libqt6printsupport6:amd64 (6.10.2+dfsg-7) ...
Setting up tshark (4.6.4-1) ...
Setting up libqt6quick6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6multimedia6:amd64 (6.10.2-2) ...
Setting up wireshark (4.6.4-1) ...
Setting up libqt6waylandcompositor6:amd64 (6.10.2-4) ...
Setting up qt6-wayland:amd64 (6.10.2-4) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libselinux1:amd64 (3.9-4build1) ...
Processing triggers for shared-mime-info (2.4-5build3) ...
Processing triggers for udev (259.5-0ubuntu3) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
```

Verify TShark installation and version:

```
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
naman@DESKTOP-PQFNDR1:~$ tshark --version
TShark (Wireshark) 4.6.4.

Copyright 1998-2026 Gerald Combs <gerald@wireshark.org> and contributors.
Licensed under the terms of the GNU General Public License (version 2 or later).
This is free software; see the file named COPYING in the distribution. There is
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compile-time info:
 Bit width: 64-bit
  Compiler: GCC 15.2.0
      GLib: 2.87.3
 With:
  +brotli                      +MaxMind
  +Gcrypt 1.12.0               +nghttp2 1.68.0
  +GnuTLS 3.8.12 and PKCS#11   +nghttp3 1.12.0
  +Kerberos (MIT)              +PCRE2 10.46 2025-08-27
  +libnl 3                     +POSIX capabilities (Linux)
  +libpcap                     +Snappy 1.2.2
  +libsmi 0.4.8                +xxhash 0.8.3
  +libxml2 2.15.1              +zlib 1.3.1
  +Lua 5.4.8                   +Zstandard 1.5.7
  +LZ4 1.10.0
 Without:
  -zlib-ng

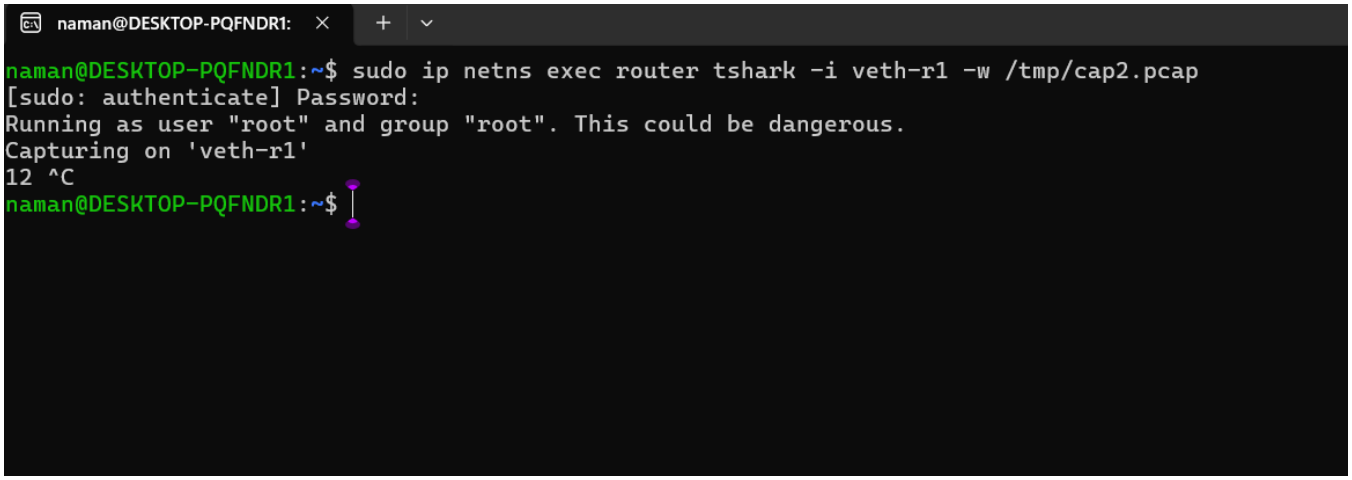
Runtime info:
      OS: Linux 6.18.33.2-microsoft-standard-WSL2
     CPU: 12th Gen Intel(R) Core(TM) i5-1235U (with SSE4.2)
  Memory: 3758 MB of physical memory
    GLib: 2.88.0
  Locale: LC_TYPE=C.UTF-8
 Plugins: supported, 0 loaded
 With:
  +brotli 1.2.0
  +c-ares 1.34.6
  +Gcrypt 1.12.0
  +GnuTLS 3.8.12
  +libpcap 1.10.6 (64-bit time_t, with TPACKET_V3)
  +libsmi 0.4.8
  +LZ4 1.10.0
  +nghttp2 1.68.0
  +nghttp3 1.12.0
  +PCRE2 10.46 2025-08-27
  +xxhash 803
  +zlib 1.3.1
  +Zstandard 1.5.7
```

Create namespaces, connect interfaces, configure IP addresses, routing and ping:

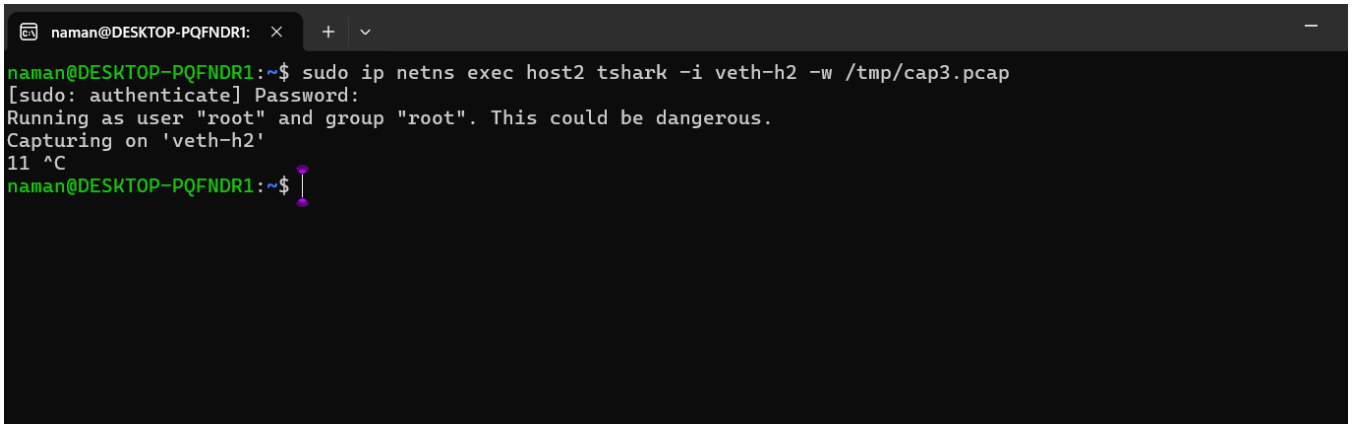
```
naman@DESKTOP-PQFNDR1:~$ sudo ip netns list
naman@DESKTOP-PQFNDR1:~$ sudo ip netns add host1
naman@DESKTOP-PQFNDR1:~$ sudo ip netns add router
naman@DESKTOP-PQFNDR1:~$ sudo ip netns add host2
naman@DESKTOP-PQFNDR1:~$ sudo ip link add veth-h1 type veth peer name veth-r1
naman@DESKTOP-PQFNDR1:~$ sudo ip link add veth-r2 type veth peer name veth-h2
naman@DESKTOP-PQFNDR1:~$ sudo ip link set veth-h1 netns host1
naman@DESKTOP-PQFNDR1:~$ sudo ip link set veth-r1 netns router
naman@DESKTOP-PQFNDR1:~$ sudo ip link set veth-r2 netns router
naman@DESKTOP-PQFNDR1:~$ sudo ip link set veth-h2 netns host2
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec host1 ip link set veth-h1 up
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec router ip link set veth-r1 up
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec router ip link set veth-r2 up
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec host2 ip link set veth-h2 up
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec host1 ip link set lo up
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec router ip link set lo up
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec host2 ip link set lo up
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec host1 ping -c 4 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.475 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.076 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.108 ms
64 bytes from 10.0.2.2: icmp_seq=4 ttl=63 time=0.089 ms

--- 10.0.2.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3062ms
rtt min/avg/max/mdev = 0.076/0.187/0.475/0.166 ms
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
naman@DESKTOP-PQFNDR1:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
13 ^C
naman@DESKTOP-PQFNDR1:~$
```

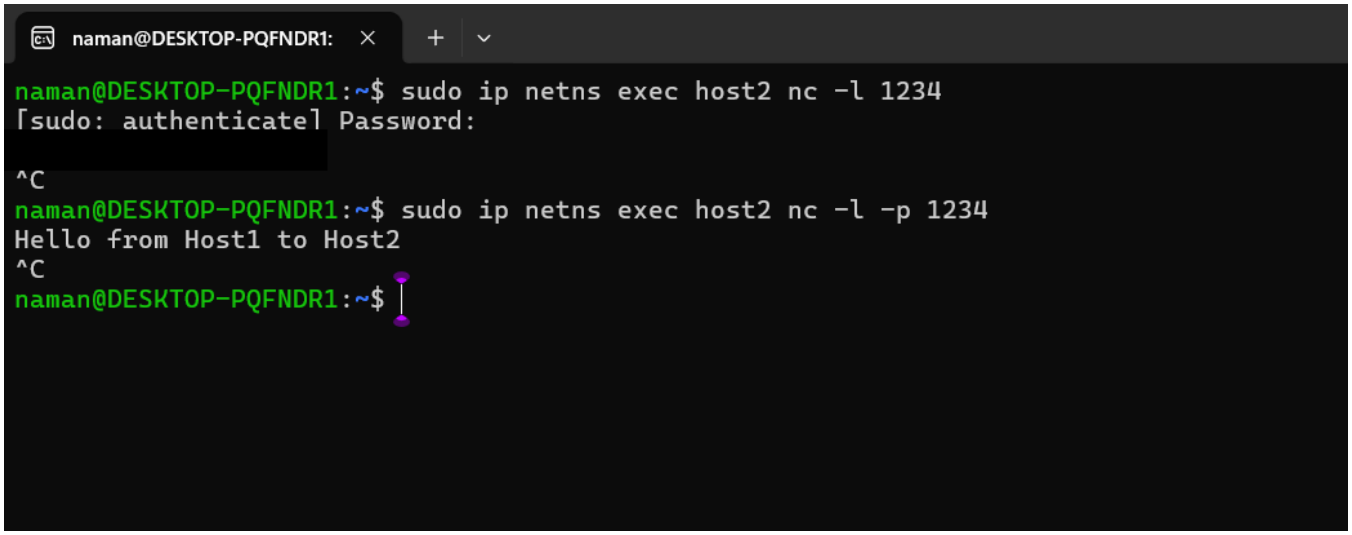
Router capture – cap2.pcap:



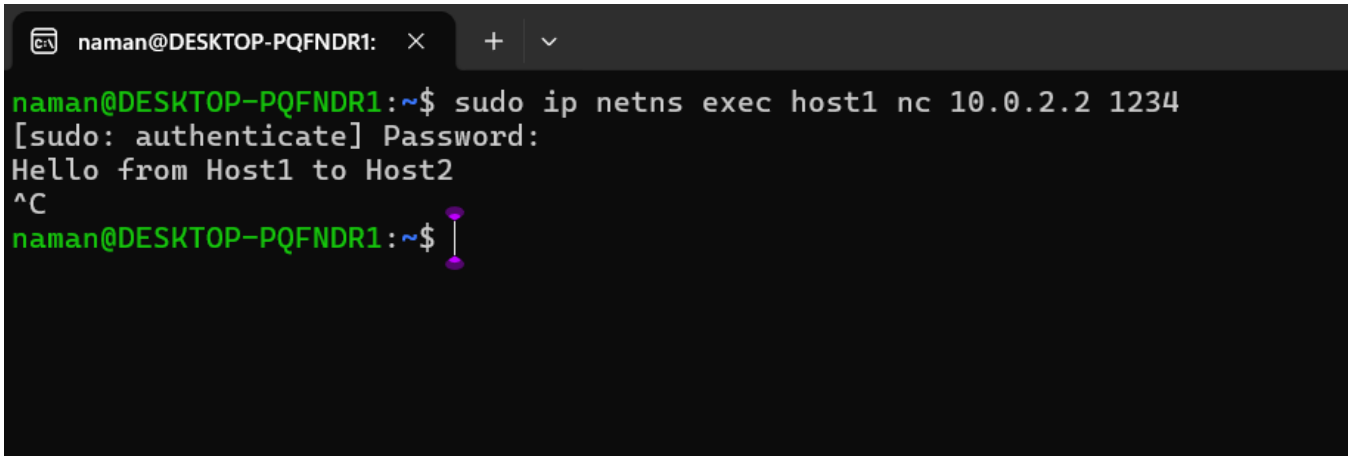
Host2 capture – cap3.pcap:



Host2 receiver – message received:



Host1 sender – message sent:



Verify PCAP files and run analysis:

```
naman@DESKTOP-PQFNDR1:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
[sudo: authenticate] Password:
-rw-r--r-- 1 root root 1.6K Nov 12 17:41 /tmp/cap1.pcap
-rw-r--r-- 1 root root 1.5K Nov 12 17:41 /tmp/cap2.pcap
-rw-r--r-- 1 root root 1.4K Nov 12 17:41 /tmp/cap3.pcap
naman@DESKTOP-PQFNDR1:~$ echo "========== 1. HOST1 PACKETS =========="
tshark -r /tmp/cap1.pcap -q -z io,phs

echo "========== 2. ROUTER PACKETS =========="
tshark -r /tmp/cap2.pcap -q -z io,phs

echo "========== 3. HOST2 PACKETS =========="
tshark -r /tmp/cap3.pcap -q -z io,phs

echo "========== 4. TTL AT ALL THREE POINTS =========="
for f in /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap; do
    echo "--- $f ---"
    tshark -r "$f" -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
done

echo "========== 5. TCP RETRANSMISSIONS =========="
for f in /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap; do
    echo -n "$f: "
    tshark -r "$f" -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
done

echo "========== 6. MESSAGE DATA =========="
tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234 && tcp.payload" -T fields -e tcp.payload 2>/dev/null

echo "========== 7. PACKET SUMMARY =========="
for f in /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap; do
    echo "--- $f ---"
    tshark -r "$f" -q -z conv,tcp
done
========== 1. HOST1 PACKETS ==========

================================================================
Protocol Hierarchy Statistics
Filter:
```

Protocol hierarchy statistics:

```
========== 1. HOST1 PACKETS ==========

================================================================
Protocol Hierarchy Statistics
Filter:

frame                                    frames:13 bytes:820
  eth                                    frames:13 bytes:820
    ipv6                                 frames:4 bytes:280
      icmpv6                             frames:4 bytes:280
    ip                                   frames:5 bytes:372
      tcp                                frames:5 bytes:372
        data                             frames:1 bytes:92
    arp                                  frames:4 bytes:168
================================================================
========== 2. ROUTER PACKETS ==========

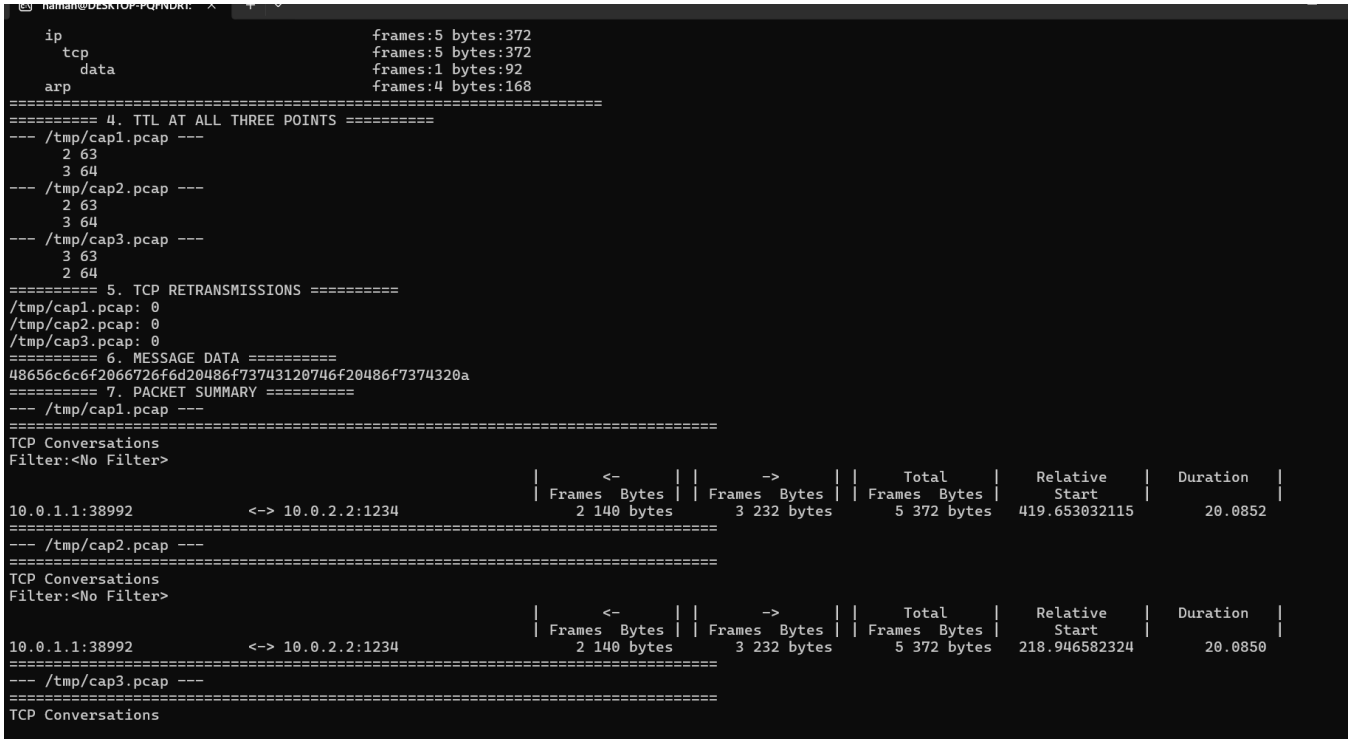
================================================================
Protocol Hierarchy Statistics
Filter:

frame                                    frames:12 bytes:750
  eth                                    frames:12 bytes:750
    ipv6                                 frames:3 bytes:210
      icmpv6                             frames:3 bytes:210
    ip                                   frames:5 bytes:372
      tcp                                frames:5 bytes:372
        data                             frames:1 bytes:92
    arp                                  frames:4 bytes:168
================================================================
========== 3. HOST2 PACKETS ==========

================================================================
Protocol Hierarchy Statistics
Filter:

frame                                    frames:11 bytes:680
  eth                                    frames:11 bytes:680
    ipv6                                 frames:2 bytes:140
      icmpv6                             frames:2 bytes:140
    ip                                   frames:5 bytes:372
```

TTL, retransmissions, payload and TCP conversation:

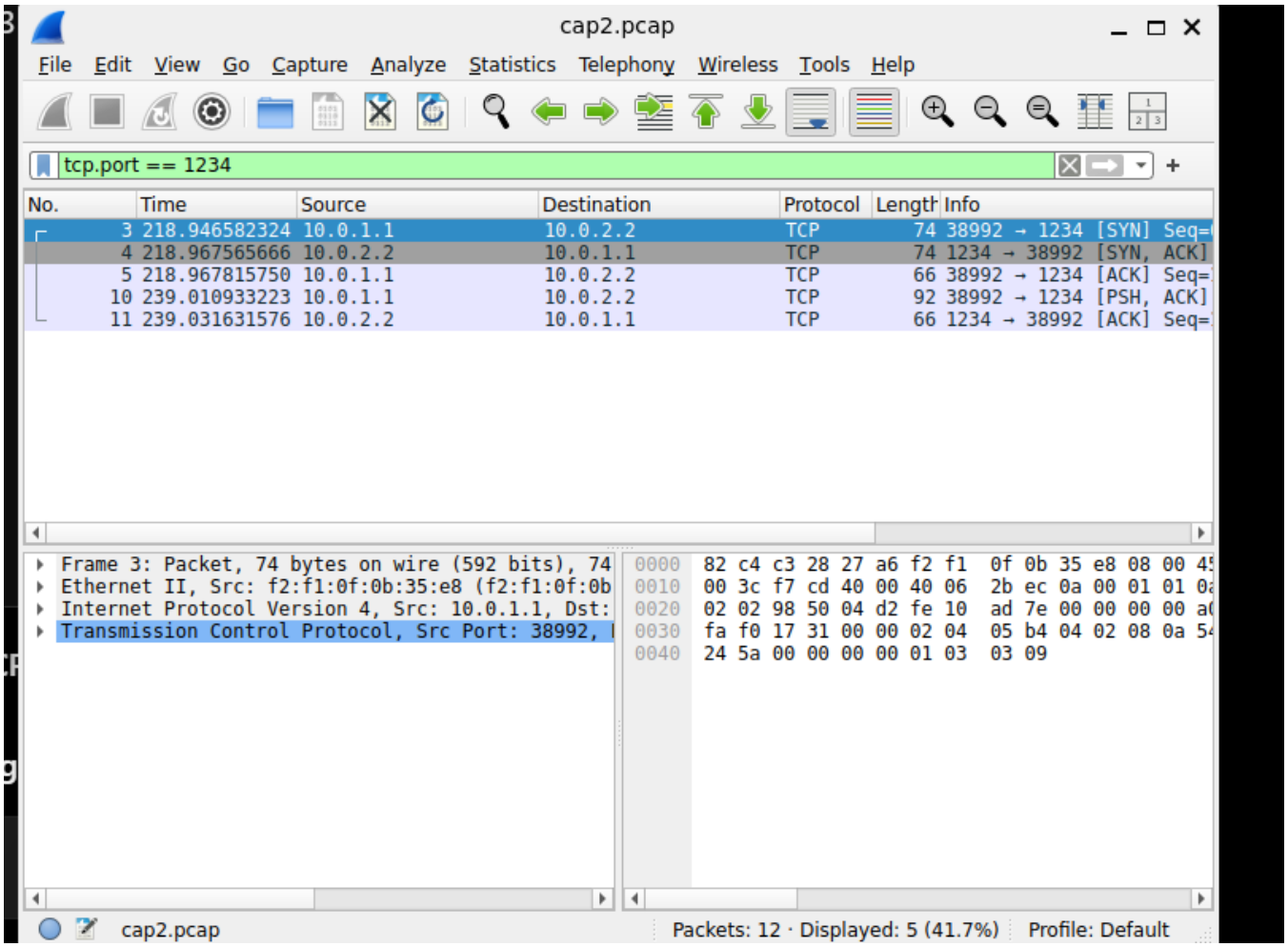


```
    ip                                  frames:5 bytes:372
      tcp                               frames:5 bytes:372
        data                            frames:1 bytes:92
    arp                                 frames:4 bytes:168
===================================================================
========== 4. TTL AT ALL THREE POINTS ==========
--- /tmp/cap1.pcap ---
      2 63
      3 64
--- /tmp/cap2.pcap ---
      2 63
      3 64
--- /tmp/cap3.pcap ---
      3 63
      2 64
========== 5. TCP RETRANSMISSIONS ==========
/tmp/cap1.pcap: 0
/tmp/cap2.pcap: 0
/tmp/cap3.pcap: 0
========== 6. MESSAGE DATA ==========
48656c6c6f2066726f6d20486f73743120746f20486f7374320a
========== 7. PACKET SUMMARY ==========
--- /tmp/cap1.pcap ---
================================================================================
TCP Conversations
Filter:<No Filter>
                                                     |       <-      | |       ->      | |     Total     |    Relative    |   Duration   |
                                                     | Frames  Bytes | | Frames  Bytes | | Frames  Bytes |      Start     |              |
10.0.1.1:38992             <-> 10.0.2.2:1234             2 140 bytes       3 232 bytes       5 372 bytes   419.653032115        20.0852
================================================================================
--- /tmp/cap2.pcap ---
================================================================================
TCP Conversations
Filter:<No Filter>
                                                     |       <-      | |       ->      | |     Total     |    Relative    |   Duration   |
                                                     | Frames  Bytes | | Frames  Bytes | | Frames  Bytes |      Start     |              |
10.0.1.1:38992             <-> 10.0.2.2:1234             2 140 bytes       3 232 bytes       5 372 bytes   218.946582324        20.0850
================================================================================
--- /tmp/cap3.pcap ---
================================================================================
TCP Conversations
```

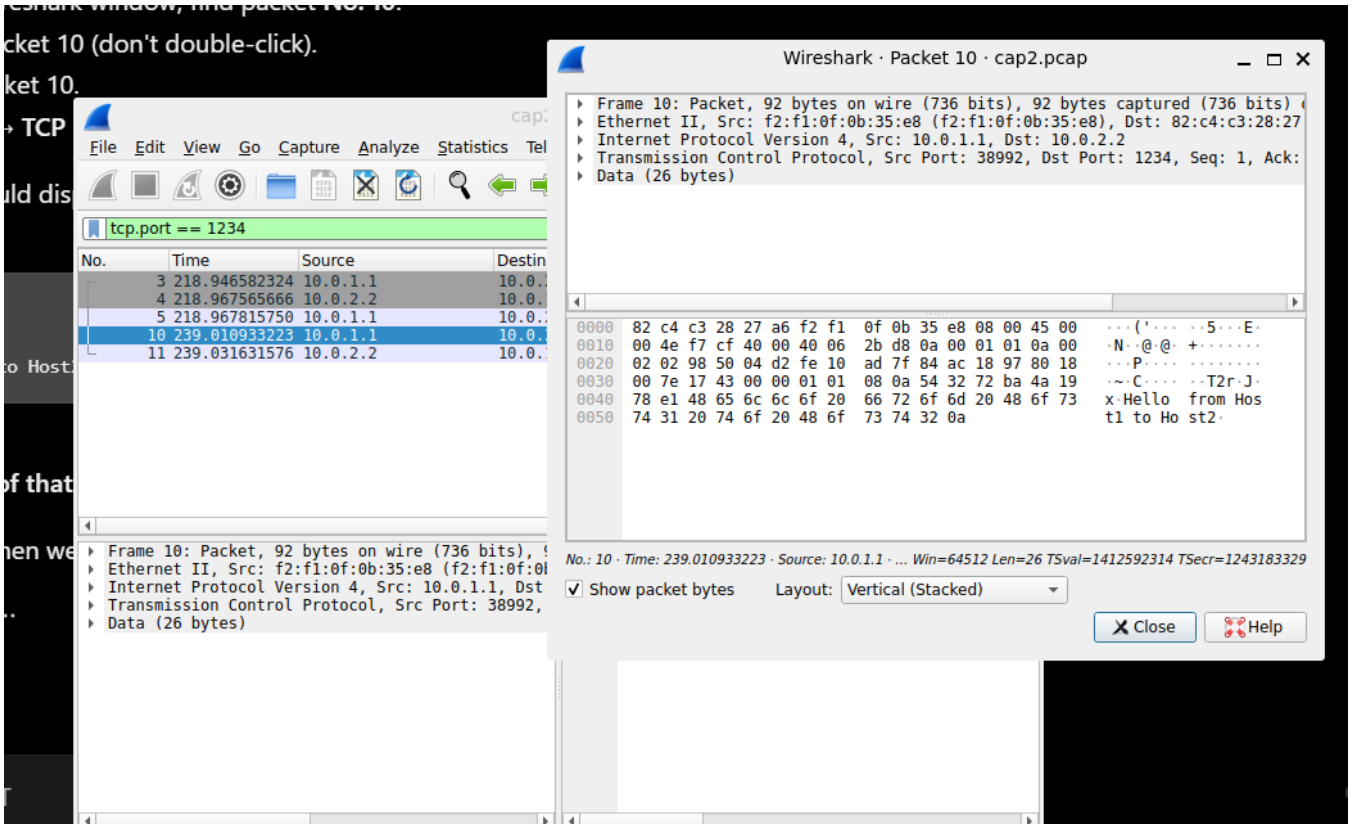
TCP SYN timestamps and Wireshark commands:

```
48656c6c6f2066726f6d20486f737431207 46f20486f7374320a
========== 7. PACKET SUMMARY ==========
--- /tmp/cap1.pcap ---
================================================================================
TCP Conversations
Filter:<No Filter>
                                                    |       <-      | |       ->      | |     Total     |    Relative    |   Duration   |
                                                    | Frames  Bytes | | Frames  Bytes | | Frames  Bytes |      Start     |              |
10.0.1.1:38992             <-> 10.0.2.2:1234            2 140 bytes       3 232 bytes       5 372 bytes   419.653032115        20.0852
================================================================================
--- /tmp/cap2.pcap ---
================================================================================
TCP Conversations
Filter:<No Filter>
                                                    |       <-      | |       ->      | |     Total     |    Relative    |   Duration   |
                                                    | Frames  Bytes | | Frames  Bytes | | Frames  Bytes |      Start     |              |
10.0.1.1:38992             <-> 10.0.2.2:1234            2 140 bytes       3 232 bytes       5 372 bytes   218.946582324        20.0850
================================================================================
--- /tmp/cap3.pcap ---
================================================================================
TCP Conversations
Filter:<No Filter>
                                                    |       <-      | |       ->      | |     Total     |    Relative    |   Duration   |
                                                    | Frames  Bytes | | Frames  Bytes | | Frames  Bytes |      Start     |              |
10.0.1.1:38992             <-> 10.0.2.2:1234            2 140 bytes       3 232 bytes       5 372 bytes   104.281555705        20.0644
================================================================================
naman@DESKTOP-PQFNDR1:~$ for f in /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap; do
    echo "========== $f =========="
    tshark -r "$f" -Y "tcp.flags.syn == 1" \
    -T fields -e frame.time_epoch -e ip.src -e ip.dst -e tcp.flags.ack
done
========== /tmp/cap1.pcap ==========
1794505086.620147084    10.0.1.1        10.0.2.2        False
1794505086.641208297    10.0.2.2        10.0.1.1        True
========== /tmp/cap2.pcap ==========
1794505086.620220814    10.0.1.1        10.0.2.2        False
1794505086.641204156    10.0.2.2        10.0.1.1        True
========== /tmp/cap3.pcap ==========
1794505086.640710306    10.0.1.1        10.0.2.2        False
1794505086.641178826    10.0.2.2        10.0.1.1        True
naman@DESKTOP-PQFNDR1:~$ wireshark /tmp/cap2.pcap
^C
naman@DESKTOP-PQFNDR1:~$ wireshark /tmp/cap3.pcap
naman@DESKTOP-PQFNDR1:~$
```

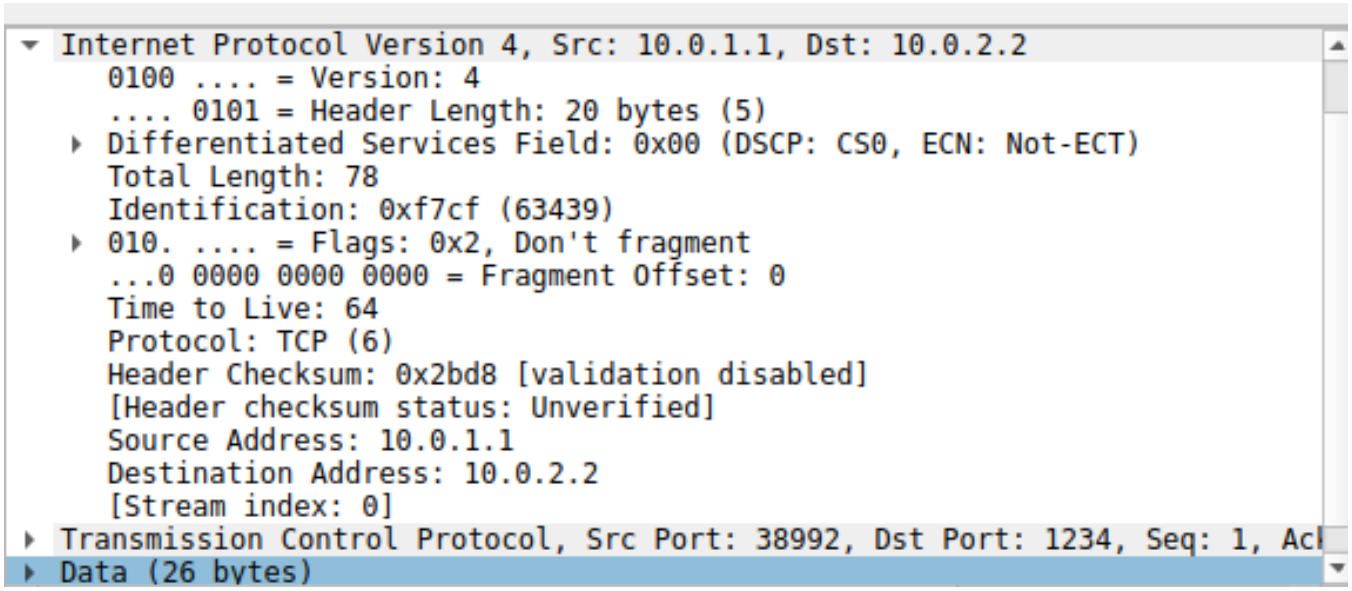
Wireshark – router TCP packets:



Wireshark – message payload in packet 10:



Wireshark – TTL 64 at router input:



Wireshark – TTL 63 at Host2:

